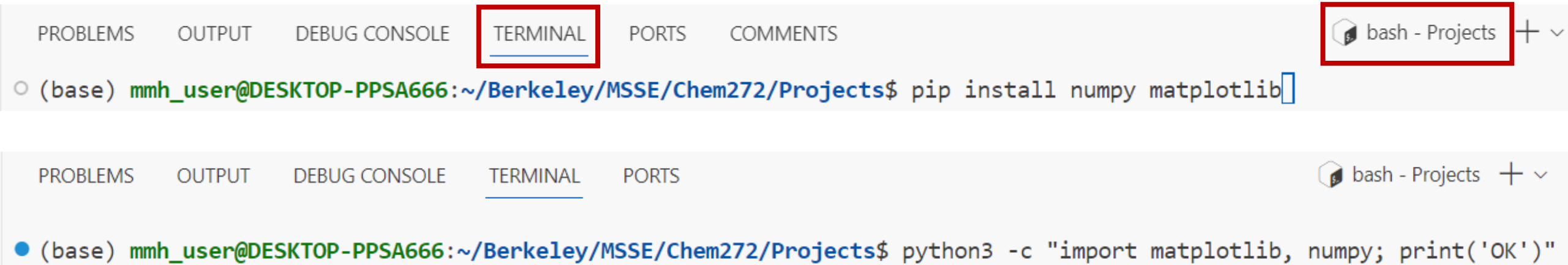


This is an installation guide for matplotlibcpp in order to generate plots in cpp using the syntax of matplotlib

1) Open VS code

2) Navigate to the directory of your project

3) make sure you have numpy and matplotlib installed by running the following commands in your VS code terminal



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS bash - Projects + v
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects$ pip install numpy matplotlib

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - Projects + v
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects$ python3 -c "import matplotlib, numpy; print('OK')"
```

It should return “OK”

4) Download the matplotlibcpp header by running the following line in you VS Code terminal

```
wget https://raw.githubusercontent.com/lava/matplotlib-cpp/master/matplotlibcpp.h
```

You should see the matplotlibcpp header in your directory now:

```
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects$ ls -lrt
total 108
drwxr-xr-x 4 mmh_user mmh_user 4096 Jul 11 02:44 'Project 2'
-rw-r--r-- 1 mmh_user mmh_user 99741 Jul 30 23:16 matplotlibcpp.h
-rw-r--r-- 1 mmh_user mmh_user 322 Jul 30 23:17 test.cpp
```

5) Generate a test2.cpp script. Note, that I used namespace. Instead of plt. in Python, you can now use the syntax plt::

```
#include <matplotlibcpp.h>
```

```
namespace plt = matplotlibcpp;
```

```
int main() {
```

```
    std::vector<double> x = {1, 2, 3, 4};
    std::vector<double> y = {1, 4, 9, 16};
```

```
    plt::plot(x, y);
    plt::title("My first C++ plot");
    plt::xlabel("x");
    plt::ylabel("y");
    plt::show();
```

```
    return 0;
```

```
}
```

6) Compile the code in VS Code. Make sure that `test.cpp` and `matplotlibcpp.h` are located in the same directory:

```
g++ -std=c++11 test2.cpp\
```

```
-I. \python3-config --includes` \  
-I$(python3 -c "import numpy; print(numpy.get_include())") \  
`python3-config --ldflags` -lpython3.9\  
-o my_plot_example
```

pointing to python config files

Your python version!

```
○ (base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects$ g++ -std=c++11 test2.cpp \  
-I. \  
`python3-config --includes` \  
-I$(python3 -c "import numpy; print(numpy.get_include())") \  
`python3-config --ldflags` -lpython3.9 \  
-o my_plot_example
```

7) Run the executable `./my_plot_example` and you should see the plot

```
5  int main() {
```

```
6
```

```
7
```

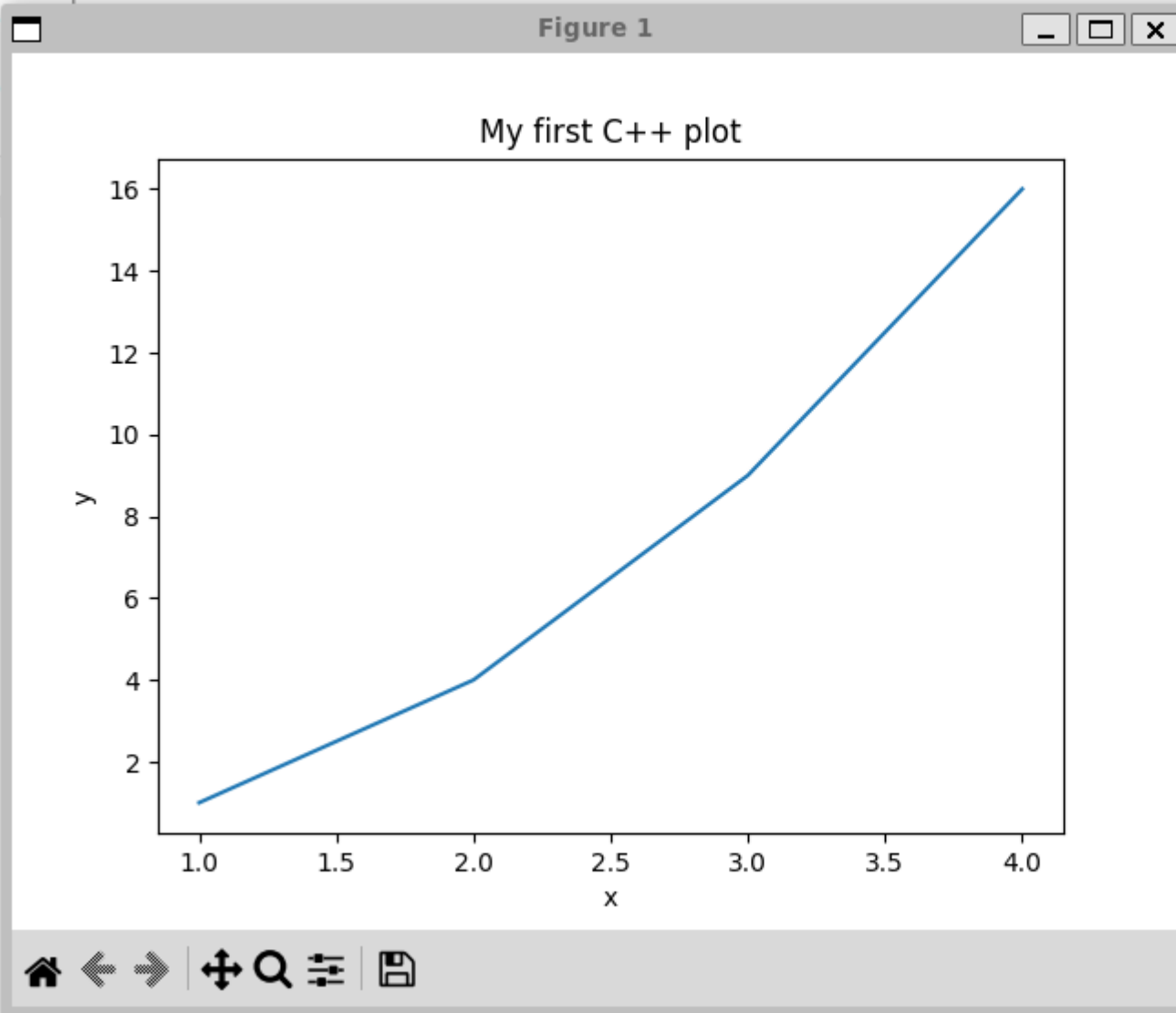
```
8
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```



```
ts$ g++ -std=c++11 test2.cp
```

```
ts$ ./my_plot_example
```